

1. (12 pts)

(a) (3 pts) Consider $x^2 - 4y^2 - z = -9$.

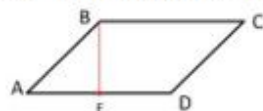
- Give the 2D name of the traces when $x = k$ is fixed.

Name: PARABOLAS

- Give the precise name of the 3D shape given by $x^2 - 4y^2 - z = -9$

Name: HYPERBOLIC PARABOLOID

(b) Consider the parallelogram show with $A(1, 1, 2)$, $B(2, 3, 7)$, $C(5, 3, 11)$, $D(4, 1, 6)$.



i. (4 pts) Find the area of the parallelogram ABCD.

$$\begin{aligned} \vec{AB} &= \langle 1, 2, 5 \rangle \\ \vec{AD} &= \langle 3, 0, 4 \rangle \\ &= (8-0)\vec{i} - (4-19)\vec{j} + (0-6)\vec{k} \\ &= \langle 8, 11, -6 \rangle \end{aligned}$$

$$\begin{aligned} \text{CHECK: } 8 + 22 - 36 &= 0 \checkmark \\ 24 + 0 - 24 &= 0 \checkmark \end{aligned}$$

$$\begin{aligned} \text{Area} &= |\langle 8, 11, -6 \rangle| \\ &= \sqrt{64 + 121 + 36} \\ &= \sqrt{221} \\ &\approx 14.87 \end{aligned}$$

Area = $\sqrt{221}$

ii. (3 pts) Find the vector of length 5 that points in the same direction as $\vec{BD}$.

$$\vec{BD} = \langle 2, -2, -1 \rangle$$

$$\frac{1}{\sqrt{4+4+1}} \langle 2, -2, -1 \rangle \leftarrow \text{UNIT VECTOR}$$

$$\frac{5}{3} \langle 2, -2, -1 \rangle$$

VERSION 2: 8
 $\langle \frac{16}{3}, -\frac{16}{3}, -\frac{8}{3} \rangle$

Vector: $\langle \frac{16}{3}, -\frac{16}{3}, -\frac{8}{3} \rangle$

iii. (2 pts) The line segment BE (shown) is perpendicular to the segment AD . Find the length of the segment AE .

$$|AE| = \text{Comp}_{\vec{AD}} \vec{AB} = \frac{\vec{AB} \cdot \vec{AD}}{|\vec{AD}|} = \frac{3+0+26}{\sqrt{9+0+16}} = \frac{23}{5}$$

Length = $|AE| = \frac{23}{5}$

2. (14 pts)

(a) (2 pts) True or false, circle one for each statement:

VERSION B

• TRUE or FALSE : Two planes are always parallel or intersecting.

← SWAP



• TRUE or FALSE: Two planes perpendicular to a given plane must be parallel.

(b) (6 pts) Consider the line that contains the point $(5, 0, 0)$ and is orthogonal to the plane $3y - 4z = 10$. Find the two points of intersection of this line with the paraboloid $20x = y^2 + z^2$. (First find parametric equations for the line!)

$$x = 5 + 0t$$

$$y = 0 + 3t$$

$$z = 0 - 4t$$

INTERSECTION:

$$20(5) = (3t)^2 + (-4t)^2$$

$$100 = 9t^2 + 16t^2$$

$$100 = 25t^2$$

$$t^2 = 4$$

$$t = \pm 2 \rightarrow \begin{matrix} \nearrow (5, 6, -8) \\ \searrow (5, -6, 8) \end{matrix}$$

Intersection Points: $(x, y, z) = \underline{(5, 6, -8), (5, -6, 8)}$

(c) (6 pts) Find an equation for the plane that passes through the point $A(0, 0, 2)$ and contains the line of intersection of the planes $x + y - z = 1$ and $2x + y - 3z = -1$. And give the x -intercept of this new plane equation.

$$\begin{cases} \textcircled{1} x + y - z = 1 \\ \textcircled{2} 2x + y - 3z = -1 \end{cases} \rightarrow -x + 2z = 2$$

$$x=0 \Rightarrow z=1 \Rightarrow y=2 \quad B(0, 2, 1)$$

$$z=0 \Rightarrow x=-2 \Rightarrow y=3 \quad C(-2, 3, 0)$$

NOW WE HAVE 3 PTS! IT IS POSSIBLE TO FIND MANY OTHERS

$$\vec{AB} = \langle 0, 2, -1 \rangle$$

$$\vec{AC} = \langle -2, 3, -2 \rangle$$

$$\text{CHECK: } 0 + 4 - 4 = 0 \checkmark$$

$$2 + 6 - 8 = 0 \checkmark$$

$$(-4 - 3)\vec{i} - (0 - 2)\vec{j} + (0 - 4)\vec{k} = \langle -1, 2, 4 \rangle$$

$$x\text{-intercept} \Rightarrow (?, 0, 0)$$

$$\Rightarrow -x + 0 + 4(0 - 2) = 0$$

$$-x - 8 = 0 \Rightarrow x = -8$$

$$\text{Plane Equation: } -x + 2y + 4(z - 2) = 0$$

$$x\text{-intercept: } (x, y, z) = \underline{(-8, 0, 0)}$$

3. (12 pts)

(a) (6 pts) Find the angle of intersection of the curves: (Round final answer to the nearest degree)

• $\mathbf{r}_1(t) = \langle t, 3-t, t^4 \rangle$
 • $\mathbf{r}_2(u) = \langle 2-u, 2u-2, 10-u^2 \rangle$

① $t = 2-u$
 ② $3-t = 2u-2$
 ③ $t^4 = 10-u^2$

Check: $t^4 = (-1)^4 = 1$
 $10-u^2 = 10-(-1)^2 = 9$

$\mathbf{r}_1'(t) = \langle 1, -1, 4t^3 \rangle$

$\mathbf{r}_1'(-1) = \langle 1, -1, -4 \rangle$

$\mathbf{r}_2'(u) = \langle -1, 2, -2u \rangle$

$\mathbf{r}_2'(3) = \langle -1, 2, -6 \rangle$

$\cos \theta = \frac{\mathbf{u} \cdot \mathbf{v}}{|\mathbf{u}| |\mathbf{v}|} = \frac{-1-2+24}{\sqrt{1+1+16} \sqrt{1+4+36}}$

$\theta = \cos^{-1} \left(\frac{21}{\sqrt{18} \sqrt{41}} \right)$
 $\approx 39.37^\circ$

Angle = 39°

(b) (6 pts) Let C be the curve of intersection of the surface $y = \frac{1}{2}x^2$ and the surface $z = \frac{1}{2}xy$. Parameterize this curve, then use the parameterization to give the arc length the curve from the point $(0,0,0)$ to $(3, \frac{9}{2}, 18)$

$x = t$ $(0,0,0) \Rightarrow t=0$
 $y = \frac{1}{2}t^2$ $(3, \frac{9}{2}, 18) \Rightarrow t=3$
 $z = \frac{1}{6}t^3$

$\int_0^3 \sqrt{(1)^2 + (t)^2 + (\frac{1}{2}t^2)^2} dt$
 $= \int_0^3 \sqrt{1+t^2+\frac{1}{4}t^4} dt$
 $= \int_0^3 \sqrt{\frac{1}{4}(4+4t^2+t^4)} dt$
 $= \frac{1}{2} \int_0^3 \sqrt{(t^2+2)^2} dt = \frac{1}{2} \int_0^3 (t^2+2) dt = \frac{1}{2} \left(\frac{1}{3}t^3 + 2t \right) \Big|_0^3$
 $= \frac{1}{2} (9+6) = \frac{15}{2}$

Arc Length = $\frac{15}{2}$

VERSION B

$7+u-u^2 \rightarrow$

$\mathbf{r}_2'(4) = \langle -1, 2, 1-2u \rangle$

$\mathbf{r}_2'(3) = \langle -1, 2, -5 \rangle$

$\cos^{-1} \left(\frac{-1-2+26}{\sqrt{18} \sqrt{1+4+25}} \right)$

$\cos^{-1} \left(\frac{17}{\sqrt{18} \sqrt{30}} \right)$

$\approx 42.98^\circ$
 $\approx 43^\circ$

4. (12 pts)

(a) (2 pts) True or false, circle one for each statement:

VERSION B

i. TRUE or FALSE : $\mathbf{r}'(t)$ and $\mathbf{N}(t)$ are always orthogonal.

↗ SWAP

ii. TRUE or FALSE : $\mathbf{r}''(t)$ and $\mathbf{N}(t)$ are always parallel.

(b) Consider $\mathbf{r}(t) = \langle t^2, 3t + 6, -2t^2 \rangle$.

i. (5 pts) Find the curvature at $t = 0$. (Give your answer as a decimal rounded to three digits)

$$\mathbf{r}'(t) = \langle 2t, 3, -4t \rangle$$

$$\mathbf{r}''(t) = \langle 2, 0, -4 \rangle$$

$$\mathbf{r}'(0) = \langle 0, 3, 0 \rangle$$

$$\mathbf{r}''(0) = \langle 2, 0, -4 \rangle$$

$$\begin{aligned} & (-12-0)\mathbf{i} - (0-0)\mathbf{j} + (0-6)\mathbf{k} \\ & = \langle -12, 0, -6 \rangle \quad \text{CHECK: } 6+0+6=0 \checkmark \\ & \quad \quad \quad -24+0+24=0 \checkmark \end{aligned}$$

$$\begin{aligned} & \frac{|\mathbf{r}'(0) \times \mathbf{r}''(0)|}{|\mathbf{r}'(0)|^3} \\ & = \frac{\sqrt{144+0+36}}{(\sqrt{0+9+0})^3} \\ & = \frac{\sqrt{180}}{3^3} = \frac{19\sqrt{20}}{3^3} \\ & = \frac{3.2\sqrt{5}}{3^3} = \frac{2\sqrt{5}}{9} \\ & \approx 0.4969 \end{aligned}$$

$$\kappa(0) = \underline{0.497}$$

ii. (5 pts) Find the equation of the tangent line to $\mathbf{r}(t)$ at the point $(4, 12, -8)$ and find where this line intersects the xz -plane

$$x = t^2 = 4, y = 3t + 6 = 12, z = -2t^2 = -8 \Rightarrow \boxed{t=2}$$

$$\mathbf{r}'(t) = \langle 2t, 3, -4t \rangle \Rightarrow \mathbf{r}'(2) = \langle 4, 3, -8 \rangle$$

$$x = 4 + 4t$$

$$y = 12 + 3t$$

$$z = -8 - 8t$$

INTERSECT xz -PLANE

$$\Rightarrow 12 + 3t = 0 \Rightarrow t = -\frac{12}{3} = -4$$

$$\Rightarrow x = 4 - 16 = -12$$

$$\begin{aligned} y &= 0 \\ z &= -8 + 32 = 24 \end{aligned}$$

$$\text{Intersection with } xy\text{-plane: } (x, y, z) = \underline{(-12, 0, 24)}$$